ASCIIgent: Training-Free Agentic ASCII Art Generation

Modern Computer Vision — Final Project Proposal

Hana Hasan Amit Arad Roei Levi Tom Shur

Goal. ASCII art forces a model to draw with 95 characters instead of pixels. We ask whether LLMs can do this through generalizable spatial reasoning rather than memorized recall, and whether a training-free agentic loop – render the output, look at it, revise – improves generation across four settings: text-to-ASCII, conversational editing, image-to-ASCII, and animation. Every design choice below is a hypothesis, not a result.

Method. P1–P3 are the required core; P4–P6 extend the same loop to editing, image, and animation, pursued in that order.

- **P1 – Text-to-ASCII (core).** Seen/held-out category split (cat, dog, bear, horse, plus held-out classes); zero-shot vs. few-shot-with-exemplars vs. procedural prompting, to separate transferable skill from leakage.
- **P2 – Agentic self-perception loop (core). draft → render → perceive → revise**, repeated a few rounds. A model critiques a rendered bitmap of its own drawing, not the raw string, against no-feedback and text-only self-critique baselines, to isolate the effect of seeing the image.
- **P3 – Evaluation (core).** Model-as-judge on the rendered image vs. embedding similarity, calibrated against a small human-rating study.
- **P4 – Conversational editing.** Natural-language edits ("make the ears larger") applied to an existing drawing; does the model localize the edit and leave the rest of the grid alone?
- **P5 – Image-to-ASCII.** Classical tone/structure-based character mapping vs. a VLM-driven pipeline, on real input photos.
- **P6 – Animation.** Short ASCII frame sequences (a walking cycle, a progressive reveal) under the same loop; test frame-to-frame consistency.

Prior work & contribution. asciiart.eu is a static, human-made archive; it supplies example material, not a method. Our framing is inspired directly by SketchAgent (Vinker et al., CVPR 2025), which shows a frozen multimodal LLM can act as a drawing agent through prompting alone, no fine-tuning, planning a sketch stroke-by-stroke and revising it through dialogue. We take two things from it: that an LLM's prior knowledge alone can support open-ended visual generation, and that iterative, language-driven editing is a natural feature once basic generation works (our P4). VideoSketcher (Ren et al., arXiv:2602.15819, 2026) extends the same recipe to sequential, temporal sketches with a fine-tuned video-diffusion renderer, and inspires our animation pilot (P6) – though we keep ours training-free. Both operate in continuous vector or pixel space. ASCII art differs in a way that matters: it is drawn from the same finite alphabet a model reads and writes with, so what "generalizes" and what a model can "see" in its own output are open questions neither paper addresses. Two recent findings sharpen those questions for us: ASCIIBench (Luo et al., arXiv:2512.04125, 2025) found embedding similarity, an otherwise standard evaluation proxy, performs near chance at separating ASCII categories – why P3 treats evaluation as something to calibrate, not assume; and Wang et al. (arXiv:2504.01589, 2025) found VLMs judging ASCII art weight character-level text over global visual pattern – exactly the confound our P2 self-critique control is designed to catch.

Resources. No training anywhere in this plan. P1–P3 run through bulk inference calls to vision-capable LLM APIs (e.g. Claude, GPT, Gemini); P5/P6 may additionally use a pretrained, open-weight image model on a single GPU for reference images, inference only.

Challenges & feasibility. Recognizability is subjective – metrics are calibrated against a small human-rated set, not trusted directly. API calls can reflow monospace whitespace; we validate the rendering pipeline first. Given the text-bias result above, model-as-judge scores are checked against a scrambled-character control. P1–P3 fit a 4–5 person group within the timeline; P4–P6 are pursued incrementally.

References: Vinker et al., *SketchAgent*, CVPR 2025 (arXiv:2411.17673) · Ren et al., *VideoSketcher*, arXiv:2602.15819, 2026 · Luo et al., *ASCIIBench*, arXiv:2512.04125, 2025 · Wang et al., *Text Speaks Louder than Vision*, arXiv:2504.01589, 2025.